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Hyperconvexity and domains of discontinuity for surface groups in $SL_3\mathbb{C}$

BY

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THESIS [PhD candidates may use DISSERTATION here instead]

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Accessibility Statement

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Dedication

For my grandmother Martha, who shared her love of mathematics with me and many others.

Acknowledgements

Emily (integrity, knowledge, high standards)
Sam (love, support, meals, no pressure)
Family (you don't know why I wanted to do this, but that's ok)
Friends (kept me company, kept me happy)
Other math teachers and mentors
Gertrude (tiniest companion)

Preface

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List of Abbreviations

- $\mathbb{R}$ The real numbers
- $\mathbb{C}$ The complex numbers

Summary

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CHAPTER 1

Introduction

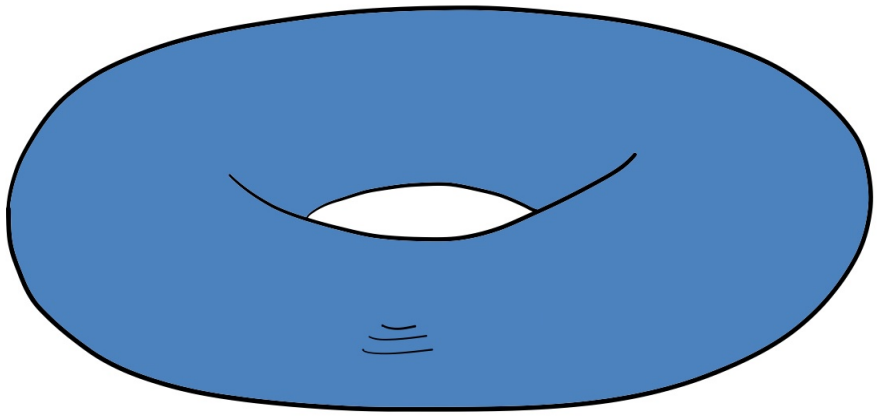

FIGURE 1. This is a torus

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & A' \arrow[r, "f'"] \arrow[d, "a"'] & B' \arrow[r, "g'"] \arrow[d, "b"'] & C' \arrow[r] \arrow[d, "c"'] & 0 \\
0 \arrow[r] & A \arrow[r, "f"] \arrow[d, "a''"'] & B \arrow[r, "g"] \arrow[d, "b''"'] & C \arrow[r] \arrow[d, "c''"'] & 0 \\
0 \arrow[r] & A'' \arrow[r, "f''"] \arrow[d] & B'' \arrow[r, "g''"] \arrow[d] & C'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

FIGURE 2. The Snake Lemma

This is the introduction chapter. We cite some classic works [**3, 4**].

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Sed porttitor urna ac ipsum rhoncus, vel dictum risus commodo. Integer ultrices dui sit amet pulvinar tempor. Phasellus nec tellus eu turpis sodales accumsan. Proin ultr

THEOREM 1.1. *This is a theorem*

We reference Theorem 1.1.

University of Illinois Chicago.

1. Motivation

1.1. Historical context. A brief overview of how the problem developed.

$$\int_0^1 f(x)dx = 2 \tag{1}$$

How to solve (1)

1.2. Open questions. Some questions remain open for future work.
Note: I have not tested the accessibility of this table.

Monkeys	Lions
100	200

Table 1. Example Table

CHAPTER 2

Background

This chapter gives necessary background.

1. Group theory

A group is a set G with a binary operation satisfying closure, associativity, identity, and inverses.

THEOREM 2.1. *Every finite subgroup of the multiplicative group of a field is cyclic.*

PROOF. This is a standard result from algebra. $\square$

CHAPTER 3

Hyperconvexity is closed under topological conjugacy

1. Richardson varieties in $\mathcal{F}^3$

TODO: This maybe belongs in an introduction / background section instead.

For any flag $x \in \mathcal{F}^3$, we define $Sch(x)$ the principle Schubert varieties associated with x. Let $x = (l, H)$ where $l \in \mathbb{CP}^2$ and $H \in (\mathbb{CP}^2)^*$ such that $l_x \subset H_x$. We distinguish the elements of $Sch(x)$ using elements of the Weyl group $W(SL_3\mathbb{C})$ as follows:

- $\alpha_{Id}(x) = \{(l, H)\} = \{x\}$,
- $\alpha_1(x) = \{(l, H_l) : l \subset H_l\}$,
- $\alpha_2(x) = \{(l_H, H) : l_H \subset H\}$,
- $\alpha_{12}(x) = \{(k, H_l) : l \subset H_l\}$,
- $\alpha_{21}(x) = \{(l_H, G) : l_H \subset H\}$, and
- $\alpha_{121}(x) = \alpha_{212}(x) = \mathcal{F}^3$.

The principle Schubert varieties are closed, and have complex dimensions $(0, 1, 1, 2, 2, 3)$, respectively.

We call a pair of flags $x, y \in \mathcal{F}^3$ *transverse* if they are in general position, meaning that for any proper, principle Schubert variety α, then $x \notin \alpha(y)$ and $y \notin \alpha(x)$.

Let $x, y \in \mathcal{F}^3$ transverse. Define $R^0_{x,y}$ to be their irreducible 0-dimensional Richardson varieties, which are the intersections between principle Schubert varieties $\alpha(x) \cap \beta(y)$ where $dim(\alpha)+dim(\beta) = dim(\mathcal{F}^3)$. There are exactly six such varieties $\{x, x_y, x^y, y_x, y^x, y\}$, where:

- $x := \{x\} \cap \alpha_{121}(y)$,
- $x_y := \alpha_1(x) \cap \alpha_{12}(y)$,
- $x^y := \alpha_2(x) \cap \alpha_{21}(y)$,
- $y_x := \alpha_{12}(x) \cap \alpha_1(y)$,
- $y^x := \alpha_{21}(x) \cap \alpha_2(y)$, and
- $y := \alpha_{121}(x) \cap \{y\}$.

Given two tranverse flags, the Richardson 0-varieties can be computed explicitly:

Example 3.1. *Choose x and y are the attracting and repelling flags of $\rho(g) = diag(\lambda, \mu, \nu)$ where $|\lambda| > |\mu| > |\nu|$. Then the six Richardson 0-varieties in $R^0_{x,y}$ are $x = (e_1, e_1 + e_2)$, $x_y = (e_1, e_1+e_3)$, $x^y = (e_2, e_1+e_2)$, $y_x = (e_3, e_1+e_3)$, $y^x = (e_2, e_2+e_3)$, and $y = (e_3, e_2+e_3)$.*

Let $x, y \in \mathcal{F}^3$ transverse, and define $R^1_{x,y}$ to be irreducible 1-dimensionsal Richardson varieties. Similarly, these are intersections of principle Schubert varieties $\alpha(x) \cap \beta(y)$ where $dim(\alpha)+dim(\beta) = dim(\mathcal{F}^3)+1$. There are six such 1-varieties, each homeomorphic to $\mathbb{CP}^1$, which we distinguish using pairs of elements of $R^0_{x,y}$.

- $span(x, x_y) := \alpha_1(x) \cap \alpha_{121}(y)$,
- $span(x, x^y) := \alpha_2(x) \cap \alpha_{121}(y)$,
- $span(x_y, y_x) := \alpha_{12}(x) \cap \alpha_{12}(y)$,
- $span(x^y, y^x) := \alpha_{21}(x) \cap \alpha_{21}(y)$,
- $span(y_x, y) := \alpha_{121}(x) \cap \alpha_1(y)$, and
- $span(y^x, y) := \alpha_{121}(x) \cap \alpha_2(y)$.

The 1-varieties are not pairwise disjoint, e.g. $\{x\} = span(x, x_y) \cap span(x, x^y)$. The points of intersection between 1-dimensional Richardson varieties are the 0-dimensional Richardson varieties.

Remark 3.1. *If $x, y \in \mathcal{F}^3$ transverse and α, β principle Schubert varieties such that $dim(\alpha) + dim(\beta) < dim(\mathcal{F}^3)$, then $\alpha(x) \cap \beta(y) =$.*

2. Assumptions and notation

Let Γ the fundamental group of a closed hyperbolic surface. Let $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$ Anosov. We use the following notation related to ρ and ρ'.

- $\xi, \xi' : \partial\Gamma \to \mathcal{F}^3$ refers to the continuous, transverse, ρ- or ρ'- equivariant, dynamics-preserving boundary maps.
- $\Lambda, \Lambda' \subset \mathcal{F}^3$ refers to the limit curves, which are the images of ξ, ξ'.
- $Th(x) = \alpha_1(x) \cup \alpha_2(x)$ refers to the balanced ideal thickening of $x \in \mathcal{F}^3$.
- $Th(\Lambda) = \cup_{x\in\Lambda} Th(x)$ refers to the thickening of the limit curve.

We call ρ, ρ' *topologically conjugate* if there exists a homeomorphism $F : \mathcal{F}^3 \to \mathcal{F}^3$ such that for all $g \in \Gamma$ and all $x \in \mathcal{F}^3$, then $F(\rho(g) \cdot x) = \rho'(g) \cdot F(x)$. We call F the *topological conjugacy* between ρ and ρ'.

Recall that ρ is *hyperconvex* if for all $x = (l_x, H_x), y = (l_y, H_y), z = (l_z, H_z) \in \Lambda$, then $l_z \not\subset l_x + l_y$ and $(H_x \cap H_y) \not\subset H_z$.

3. Hyperconvexity is closed under topological conjugacy

THEOREM 3.1. *Hyperconvexity is closed under topological conjugacy. In other words, if ρ and ρ' are topologically conjugate, then ρ is hyperconvex if and only if ρ' is hyperconvex.*

We use two supporting lemmas to prove this theorem. First, we show that the relative position of 0-dimensional Richardson varieties of Λ determines whether ρ is hyperconvex.

LEMMA 3.1.1. *An Anosov representation ρ is hyperconvex* if and only if *for all $x, y, z \in \Lambda^{(3)}$ we have $R^0_{x,y} \cap R^0_{x,z} = \{x\}$.*

For the second lemma, assume that $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between ρ and ρ'. Clearly $F(\Lambda) = \Lambda'$. This mapping is pointwise, i.e. for all $\alpha \in \partial\Gamma$, then $F(\xi(\alpha)) = \xi'(\alpha)$. So for $x \in \Lambda$, we can use the notation $x' := F(x) \in \Lambda'$.

Topological conjugacies may preserve the 0- and 1-dimensional Richardson varieties of Λ in one of two ways. We say F *matches* Richardson varieties if for all $x, y \in \Lambda^{(2)}$, then $F(x_y) = x'_{y'}$ and $F(span(x, x_y)) = span(x, x'_{y'})$. We say F *switches* Richardson varieties if for all $x, y \in \Lambda^{(2)}$, then $F(x_y) = x'^{y'}$ and $F(span(x, x_y)) = span(x, x'^{y'})$.

LEMMA 3.1.2. *For all $x, y \in \Lambda^{(2)}$, then $F(R^0_{x,y}) = R^0_{x',y'}$ and $F(R^1_{x,y}) = R^1_{x',y'}$. Furthermore, F either matches or switches Richardson varieties.*

We also have the following consequence of Lemma 3.1.2, which is of independent interest:

COROLLARY 3.1.1. *For all $x \in \Lambda$, then $F(Th(x)) = Th(x')$.*

4. Proof of Theorem 3.1.

We first prove our main theorem using our supporting lemmas.

PROOF. (Theorem 3.1) Suppose $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between Anosov representations $\rho, \rho' :\to SL_3\mathbb{C}$.

We begin with the forward direction. Assume that ρ is hyperconvex. Let $x, y, z \in \Lambda^{(3)}$. By Lemma 3.1.1, then $x_y \neq x_z$ and $x^y \neq x^z$. By Lemma 3.1.2, then the sets

$$\{F(x_y), F(x^y), F(x_z), F(x^z)\} = \{x'_{y'}, x'^{y'}, x'_{z'}, x'^{z'}\} \tag{2}$$

contain four distinct points, and in particular $x'_{y'} \neq x'_{z'}$ and $x'^{y'} \neq x'^{z'}$. Since x, y, z were general, then this holds for all triples in $x', y', z' \in \Lambda'$, and ρ' is hyperconvex.

The other direction proceeds similarly. Assume that ρ is *not* hyperconvex. By Lemma 3.1.1, there exist $x, y, z \in \Lambda^{(3)}$ such that $x_y = x_z$ or $x^y = x^z$. By Lemma 3.1.2, then F either matches or switches Richardson varieties. So then $x'_{y'} = x'_{z'}$ or $x'^{y'} = x'^{z'}$. Therefore ρ' is *not* hyperconvex. □

We also prove our lemma linking Richardson 0-dimensional varieties to hyperconvexity.

Proof. (Lemma 3.1.1) Suppose ρ is Anosov. Let $x = (l_x, H_x), y = (l_y, H_y), z = (l_z, H_z) \in \Lambda^{(3)}$. Since x, y, z are pairwise transverse, then

$$y_x, y^x, z_x, z^x \notin R^0_{x,y} \cap R^0_{x,z}. \tag{3}$$

It suffices to check the pairs $x_y, x_z \in \alpha_1(x)$ and $x^y, x^z \in \alpha_2(x)$.

By definition, $x_y = (l_x, l_x + l_y)$ and $x_z = (l_x, l_x + l_z)$, which are distinct if and only if $l_x + l_y \neq l_x + l_z$, or equivalently if and only if $l_z \not\subset l_x + l_y$.

Similarly, $x^y = (H_x \cap H_y, H_x)$ and $x^z = (H_x \cap H_z, H_x)$, which are distinct if and only if $H_x \cap H_y \neq H_x \cap H_z$, or equivalently if and only if $H_x \cap H_y \not\subset H_z$.

So $R^0_{x,y} \cap R^0_{x,z} = \{x\}$ for all $x, y, z \in \Lambda^{(3)}$ is equivalent to ρ hyperconvex. □

5. Action of $\rho(g)$ on $\mathcal{F}^3$

The remainder of this chapter is devoted to the proof of Lemma 3.1.2.

In this section we describe the action of $\rho(g)$ on $\mathcal{F}^3$ for $g \in \Gamma$ infinite order. Since ρ is Anosov, then $\rho(g)$ is strongly loxodromic, i.e. its eigenvalues have distinct magnitude. That means $\rho(g)$ has attracting and repelling limit curve points $x = \xi(g^+)$ and $y = \xi(g^-)$, respectively.

Under the action $\rho(g)$, most of $\mathcal{F}^3$ is attracted to x and repelled by y. We define the *uphill* of x as the subset attracted to x under $\rho(g)$:

$$U_{\rho(g)}(x) := \{f \in F^3 : \rho(g^n) \cdot f \to x\}. \tag{4}$$

Similarly, we define the *downhill* of y as the subset repelled by y under $\rho(g)$, or equivalently the subset attracted by y under $\rho(g^{-1})$:

$$D_{\rho(g)}(y) = U_{rho(g^{-1})}(y) := \{f \in \mathcal{F}^3 : \rho(g^{-n}) \cdot f \to y\}. \tag{5}$$

We then define the *exceptional set* of $\rho(g)$ as the subset *not* attracted to x or *not* repelled by y under $\rho(g)$:

$$E_{\rho(g)} = \mathcal{F}^3 \setminus (U_{\rho(g)}(x) \cap D_{\rho(g)}(y)). \tag{6}$$

The set $E_{\rho(g)}$ is nonempty and closed, since it is complement of an open set. It contains the fixed points of $\rho(g) \cdot \mathcal{F}^3$, which are precisely $R^0_{x,y}$. It suffices to show this for $x, y, \rho(g)$ as in Example 3.1, since up to conjugation $\rho(g) = diag(\lambda, \mu, \nu)$.

We can partition $E_{\rho(g)}$ using each element's attracting and repelling fixed points. By taking the closure of these subsets, we recover the 1-dimensional Richardson varieties $R^1_{x,y}$.

- $span(x, x_y) = \alpha_1(x) = \overline{U(x) \cap D(x_y)}$
- $span(x, x^y) = \alpha_2(x) = \overline{U(x) \cap D(x^y)}$
- $span(x_y, y_x) = \overline{U(x_y) \cap D(y_x)}$
- $span(x^y, y^x) = \overline{U(x^y) \cap D(y^x)}$

- $span(y_x, y) = \alpha_1(y) = \overline{U(y_x) \cap D(y)}$
- $span(y^x, y) = \alpha_2(y) = \overline{U(y^x) \cap D(y)}$

So $E_{\rho(g)}$ is homeomorphic to the union of six copies of $\mathbb{CP}^1$, each identified with two neighbors in a cyclic order $(x, x_y, y_x, y, y^x, x^y)$.

6. F preserves the Richardson 1-varieties of pairs $\xi(g^+), \xi(g^-)$

Assume $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$. We use the notation $x' = F(x), y' = F(y)$.

First, we simplify the conclusions that we need to check.

Proposition 3.1. *Suppose that F a topological conjugacy such that for some $x, y \in \Lambda^{(2)}$ we have F preserves the Richardson 1-varieties of x and y, i.e. $F(R^1_{x,y}) = R^1_{x',y'}$. Then*

- *F preserves the Richardson 0-varieties of x and y, i.e. $F(R^0_{x,y}) = R^0_{x',y'}$.*
- *F either matches or switches the Richardson varieties of x and y.*

Proof. Let F is a homeomorphism that preserves the Richardson 1-varieties of x and y. First, let $z \in R^0_{x,y}$. Then $z = U \cap V$ for distinct $U, V \in R^1_{x,y}$. So then

$$\begin{aligned} F(z) &= F(U \cap V) && (7)\\ &= F(U) \cap F(V) \text{ since } F \text{ a homeomorphism, which} && (8)\\ &= U' \cap V' \text{ where } U', V' \in R^1_{x',y'}, \text{ so} && (9)\\ &= z' \in R^0_{x',y'}. && (10) \end{aligned}$$

So $F(R^0_{x,y}) \subset R^0_{x',y'}$. We can prove the other containment by reversing the argument, so $F(R^0_{x,y}) = R^0_{x',y'}$.

Recall that $x' = F(x), y' = F(y) \in \Lambda'$. Since $x \in span(x, x_y) \cap span(x, x^y)$, then $x' \in F(span(x, x_y)) \cap F(span(x, x^y))$, so either $F(span(x, x_y)) = span(x', x'_{y'})$ or $F(span(x, x_y)) = span(x, x^y)$. In the first case, F matches the Richardson varieties of x, y and $F(x_y) = x'_{y'}$, and in the second case F switches the Richardson varieties of x, y and $F(x_y) = x'^{y'}$. □

Now, we show that F preserves Richardson 1-varieties. We first consider the case where x, y are the attracting and repelling flags of an infinite order element $g \in \Gamma$.

Proposition 3.2. *Let $x, y \in \Lambda$ such that $x = \xi(g^+)$ and $y = \xi(g^-)$. Then F preserves the Richardson 1-varieties of x and y.*

Proof. Since F is a topological conjugacy, it must preserve the exceptional set of $\rho(g)$, so $F(E_{\rho(g)}) = E_{\rho'(g)}$. Recall that the exceptional set of $\rho(g)$ is the union of the Richardson 1-varieties of x and y, that is:

$$E_{\rho(g)} = \cup_{V \in R^1_{x,y}} V. \tag{11}$$

Since $F|_{E_{\rho(g)}}$ is continuous and injective, then it must send 1-varieties to 1-varieties, so $F(R^1_{x,y}) = R^1_{x',y'}$. □

7. F preserves Richardson 1-varieties of Λ

In this section, we extend our argument from pairs $\xi(g^+), \xi(g^-)$ to all distinct pairs $\Lambda^{(2)}$. The main idea is that pairs of the form $\xi(g^+), \xi(g^-)$ are dense in Λ^2.

First, we extend Proposition 3.2 to Richardson varieties of pairs x, z where $x = \xi(g^+)$ and z is a (distinct) generic point on Λ.

PROPOSITION 3.3. *Let $g \in \Gamma$ infinite order, and define $x := \xi(g^+)$. Let $z \in \Lambda \setminus \{x\}$. Then F preserves the Richardson 1-varieties of x and z.*

We observe that Richardson 1-varieties vary continuously on transverse pairs of flags:

PROPOSITION 3.4. *Let $\{x_i\}_{i=1}, \{y_i\}_{i=1} \in (\mathcal{F}^3)^{\mathbb{N}}$ such that for all $i \in \mathbb{N}$, the pair (x_i, y_i) are transverse. Assume $x_i \to x$ and $y_i \to y$, and that the pair (x, y) is transverse. Then $lim_{i\to\infty} R^1_{x_i,y_i} = R^1_{x,y}$.*

This continuity extends to each type of Richardson variety, e.g. $lim_{i\to\infty} x_i^{y_i} = x^y$.

We use the following result from group theory:

REMARK 3.2 (TODO REFERENCE). *Let Γ hyperbolic with boundary $\partial\Gamma \equiv S^1$, and let $x, z \in \partial\Gamma$. Then there exists a quasigeodesic sequence $\{g_i\}_{i=1} \in \Gamma^{\mathbb{N}}$ such that for all $y \in \partial\Gamma$:*

$$lim_{i\to\infty} g_i \cdot y = \begin{cases} x : y = x \\ z : y \neq x \end{cases} . \tag{12, 13}$$

PROOF. (Proposition 3.3) If $z = \xi(g^-)$ then we are done by Prop 3.2, so assume not. Set $y := \xi(g^-)$.

Let $\{g_i\}_{i=1} \in \Gamma^{\mathbb{N}}$ as in Rem. 3.2, such that $\lim_{i=1} \rho(g_i) \cdot y = z$ and $lim_{i=1} \rho(g_i) \cdot x = x$. For all $i \in \mathbb{N}$, define the following:

$$x_i := \rho(g_i) \cdot x, \tag{14}$$

$$y_i := \rho(g_i) \cdot y, \tag{15}$$

$$x'_i := F(x_i), \text{ and} \tag{16}$$

$$y'_i := F(y_i). \tag{17}$$

By Prop. 3.2, then for all $i \in \mathbb{N}$, F preserves the Richardson 1-varieties of x_i and y_i. By Prop. 3.4, then $\lim_{i\to\infty} R^1_{x_i,y_i} = R^1_{x,z}$. By continuity of F, then $\lim_{i\to\infty} x'_i = x'$ and $\lim_{i\to\infty} y'_i = z'$. We apply Prop. 3.4 again to the transverse sequence (x'_i, y'_i), so then

$$F(R^1_{x,z}) = \lim_{i\to\infty} F(R^1_{x_i,y_i}) \tag{18}$$

$$= \lim_{i\to\infty} R^1_{x'_i,y'_i} \tag{19}$$

$$= R^1_{x',z'}. \tag{20}$$

□

Next, we extend Proposition 3.3 to Richardson varieties of pairs w, z where both w and z are distinct generic points of Λ.

PROPOSITION 3.5. *For all $w, z \in \Lambda^{(2)}$, then F preserves the Richardson 1-varieties of w and z.*

PROOF. If w or z is the attracting fixed point of an infinite order element $g \in \Gamma$, then we are done by Prop 3.3. Assume not.

Let $\{x_i\}_{i=1} \in \Lambda^{\mathbb{N}}$ such that each $x_i = \xi(g_i)$ is the attracting fixed point of an infinite order word, and $\lim_{i\to\infty} x_i = w$. This sequence exists because attracting fixed points are dense in Λ. Define $x_i' = F(x_i)$ for all $i \in \mathbb{N}$. By continuity of F, then $\lim_{i\to\infty} x_i' = w'$.

The proof from here proceeds similarly to that of Prop. 3.3. By Prop. 3.3, then for all $i \in \mathbb{N}$, F preserves the Richardson 1-varieties of x_i and z. By Prop. 3.4, then $\lim_{i\to\infty} R^1_{x_i,z} = R^1_{w,z}$. We apply Prop. 3.4 again to the transverse sequence (x_i', z'), so then

$$F(R^1_{w,z}) = \lim_{i\to\infty} F(R^1_{x_i,z}) \tag{21}$$
$$= \lim_{i\to\infty} R^1_{x_i',z'} \tag{22}$$
$$= R^1_{w',z'}. \tag{23}$$

□

8. F matches or switches Richardson varieties consistently

In this section, we show that F either matches all Richardson varieties of Λ or switches all Richardson varieties of Λ. We also highlight how F preserves the balanced ideal thickening of each $x \in \Lambda$.

Recall that $Th(x) = \alpha_1(x) \cup \alpha_2(x)$, and equivalently $Th(x) = span(x, x_y) \cup span(x, x^y)$ for some $y \in \Lambda$ distinct from x.

PROOF. (Corollary 3.1.1) Let $x \in \Lambda$, and let $y \in \Lambda$ distinct from x. By Prop. 3.5, F preserves the Richardson 1-varieties of x and y. By Prop. 3.1, F either matches or switches the Richardson varieties of x and y. So then

$$F(Th(x)) = F(span(x, x_y) \cup span(x, x^y)) \tag{24}$$
$$= span(x', x'_{y'}) \cup span(x', x'^{y'}) \tag{25}$$
$$= Th(x'). \tag{26}$$

□

COROLLARY 3.1.2. *If F matches (resp. switches) the Richardson varieties of some pair of $x, y \in \Lambda^{(2)}$, then F matches (resp. switches) the Richardson varieties of all such pairs.*

PROOF. Observe that

$$\alpha_1(\Lambda) = \cup_{x\in\Lambda} \alpha_1(x) \tag{27}$$
$$= \cup_{x,y\in\Lambda^{(2)}} span(x, x_y) \qquad \cong S^1 \times \mathbb{CP}^1 \subset \mathcal{F}^3. \tag{28}$$

Similarly, $\alpha_2(\Lambda) \cong S^1 \times \mathbb{CP}^1$. In particular, $\alpha_1(\Lambda)$ and $\alpha_2(\Lambda)$ are connected subsets of $\mathcal{F}^3$ whose intersection is Λ.

By Cor. 3.1.1, clearly

$$F(\alpha_1(\Lambda)) \cup F(\alpha_2(\Lambda)) = F(\alpha_1(\Lambda) \cup \alpha_2(\Lambda)) \tag{29}$$
$$= F(Th(\Lambda)) \tag{30}$$
$$= Th(\Lambda') \tag{31}$$
$$= \alpha_1(\Lambda') \cup \alpha_2(\Lambda'). \tag{32}$$

Since F is a homeomorphism which sends Λ to Λ', then either $F(\alpha_1(\Lambda)) = \alpha_1(\Lambda')$, or $F(\alpha_1(\Lambda)) = \alpha_2(\Lambda)$.

In other words, if F matches the Richardson varieties of one pair in Λ, then it matches the Richardson varieties of all such pairs. Conversely, if F switches the Richardson varieties of one pair in Λ, then it switches the Richardson varieties of all such pairs. □

This completes the proof of Lemma 3.1.2.

9. Conjectures and future steps

TODO: This should probably go in a different chapter eventually, maybe introduction or conclusions idk.

REMARK 3.3. *For distinct $x, y \in \mathcal{F}^3 \backslash \Lambda$, F will not necessarily preserve the 0-dimensional Richardson varieties.*

REMARK 3.4. *Theorem 3.1 applies to virtual hyperbolic surface groups as well, e.g. $\Gamma = \pi_1(\mathcal{O}^2)$ where $\mathcal{O}^2$ is a hyperbolic 2-orbifold.*

Conjectural picture for higher dimensions:

I suspect that topological conjugacies of Borel Anosov representations will preserve Richardson varieties up to $R^{n-2}_{x,y}$ organized into a cell decomposition with data an $(n-1)$-valent graph where vertices correspond to 0-varieties, edges correspond to 1-varieties, fully connected triples correspond to 2-varieties, etc. Matching/switching Richardson varieties will mean sending Richardson varieties to their dual.

I think balanced ideals I correspond to subgraphs of the Richardson variety graph containing exactly half of the vertices. In other words, $Th^I(x) \cap Th^I(y) =$, and

$$(Th^I(x) \cap R^0_{x,y}) \cup (Th^I(y) \cap R^0_{x,y}) = R^0_{x,y}. \tag{33}$$

However, topological conjugacies should not depend on the choice of balanced ideal, so then they determine diffeomorphisms between all types of quotient manifolds. In other words, if $F : \mathcal{F}^n \to \mathcal{F}^n$ a topological conjugacy between ρ, ρ', then:

$$F(M^I_\rho) = M^I_{\rho'} \text{ for all balanced ideals } I \subset W(SL_n\mathbb{C}). \tag{34}$$

Most importantly, I believe the conclusions of Theorem 3.1 will apply in higher rank:

CONJECTURE 3.1. *For $\rho, \rho' : \Gamma \to SL_n\mathbb{C}$ Borel Anosov and topologically conjugate. Then ρ hyperconvex if and only if ρ' hyperconvex.*

CHAPTER 4

Anosov components have one topological conjugacy class

1. Notation

TODO assumptions on ρ, ρ' Consider the collection of Anosov representations

$$\mathcal{A}_{\mathbb{R}}(\Gamma) := \{\rho : \Gamma \to SL_3\mathbb{R} \text{ Anosov}\}, \text{ and } \mathcal{A}(\Gamma) := \{\rho : \Gamma \to SL_3\mathbb{C} \text{ Anosov}\}. \tag{35}$$

Let $\rho_0 : \Gamma \to PSL_2\mathbb{R}$ Fuchsian and $\rho_i : \Gamma \to PSL_2\mathbb{C}$ quasi-Fuchsian. To create Anosov representations in $SL_3\mathbb{C}$, we can compose with an irreducible representation $\iota : PSL_2\mathbb{R} \to SL_3\mathbb{R}$:

$$\rho_F = \iota \circ \rho_0 : \Gamma \to SL_3\mathbb{R}, \text{ and } \rho_{QF} = \iota \circ \rho_i : \Gamma \to SL_3\mathbb{C}. \tag{36}$$

Similarly, (assuming that ρ_0, ρ_i lift to $SL_2\mathbb{R}$) we can compose with a reducible representation $\Delta : SL_2\mathbb{C} \to SL_3\mathbb{C}$:

$$\rho_R = \Delta \circ \rho_0 : \Gamma \to SL_3\mathbb{R}, \text{ and } \rho_{QR} = \Delta \circ \rho_i : \Gamma \to SL_3\mathbb{C}. \tag{37}$$

The representations ρ_{Fuchs} and ρ_{Red} are Anosov. We can then define the following subsets of $\mathcal{A}$:

- $SL_3\mathbb{C}$ quasi-Fuchsian representations $\mathcal{H}(\Gamma) := \{\rho \in \mathcal{A}(\Gamma) : \rho = \iota\circ$in the same Anosov component a
- Hitchin representations $\mathcal{H}(\Gamma) := \{\rho \in \mathcal{A}_{\mathbb{R}}(\Gamma) : \rho$in the same Anosov component as $\rho_0\}$
- Quasi-Hitchin representations
- $SL_3\mathbb{C}$-reducible representations
- Barbot representations
- Quasi-Barbot representations

TODO define Hitchin , Barbot $\mathcal{B}$, quasi-Hitchin $\mathcal{QH}$, quasi-Barbot $\mathcal{QB}$ TODO define domain of discontinuity

2. Anosov components have one topological conjugacy class

THEOREM 4.1. *Suppose $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$ in the same Anosov component. Suppose furthermore that ρ and ρ' are irreducible. Then ρ, ρ' are topologically conjugate.*

Combined with Theorem 3.1, we get the following corollaries:

COROLLARY 4.1.1. *All quasi-Hitchin representations are hyperconvex. All quasi-Barbot representations are* not *hyperconvex.*

Furthermore:

COROLLARY 4.1.2. *The quasi-Hitchin and quasi-Barbot Anosov components are disjoint.*

Additionally, distinct Anosov components may have the same topological conjugacy class. For example, the quasi-Barbot components in $Hom(T(p,q,r), SL_3\mathbb{C})$ are dual, so have the same conjugacy class:

PROPOSITION 4.1. *Let $\rho : \Gamma \to SL_3\mathbb{C}$ Anosov and let ρ^* its dual. Then ρ, ρ^* are topologically conjugate.*

We also describe the topological conjugacy classes of a reducible Anosov component $\mathcal{A}_{red}(\Gamma) \subsetneq QB(\Gamma)$.

THEOREM 4.2. *There are exactly three topological conjugacy classes of $\mathcal{A}_{red}(\Gamma)$:*

- *The class of semisimple representations,*
- *The class of line- or plane-reducible representations, and*

- *The class of reducible, but neither line- nor plane-reducible representations.*

Representations in the last class are topologically conjugate to the rest of their $QB(\Gamma)$ Anosov component.

3. Key ingredients for proof

In this section, we highlight prior results that are used in the proof of Theorem 4.1. First, we use the following theorem of Stecker:

THEOREM 4.3 (**??**, Thm 3.24). *Let $\rho : \Gamma \to G$ Borel Anosov. If $dim(\Lambda) \leq mbic(G) - 2$, then every non-empty cocompact domain of discontinuity in $\mathcal{F}^3$ is dense and connected, and there is a bijection:*

$$\{\textit{balanced ideals in } W(G)\} \leftrightarrow \{\textit{nonempty cocompact domains of discontinuity in } \mathcal{F}(G)\} \tag{38}$$

We tailor Stecker's theorem to our assumptions. Recall that $G = SL_3\mathbb{C}$, so $mbic(G) = 4$ and $\mathcal{F}(G) = \mathcal{F}^3$. Then $W(G) = A_2$ which has exactly one balanced ideal. In $SL_3\mathbb{C}$, all Anosov representations are Borel Anosov. Furthermore, we assume that Γ is a hyperbolic surface group, so $\partial\Gamma \cong S^1$. Then $1 \leq dim(\Lambda_\rho) \leq 2$, because ξ_ρ is an injective and continuous function of S^1. So the condition in Theorem 4.3 is met, and:

REMARK 4.1. *For Γ a hyperbolic surface group and $\rho : \Gamma \to SL_3\mathbb{C}$ Anosov, then ρ has a* unique *nonempty cocompact domain of discontinuity $\Omega_\rho \subset \mathcal{F}^3$. Furthermore, $\Omega_r ho$ is connected and dense in $\mathcal{F}^3$.*

We also use the following lemma, proved in the following section:

LEMMA 4.3.1. *For Γ a hyperbolic surface group and $\rho : \Gamma \to SL_3\mathbb{C}$ Anosov, then Ω_ρ is simply connected.*

Finally, we use the following theorem of Dumas and Sanders:

THEOREM 4.4 (**??**, Thm 5.1). *Let (P, Q) be parabolic subgroups of G, with P symmetric, and let $I \subset W(G)$ be a balanced ideal of type (P, Q). Let $\rho : \Gamma \to G$ be a P-Anosov representation. Then for any $\rho' \in \mathcal{A}(\rho, P)$, the quotient manifolds M^I_ρ and $M^I_{\rho'}$ are diffeomorphic.*

We again tailor this theorem to our assumptions that $G = SL_3\mathbb{C}$, where there is only symmetric parabolic subgroup and one balanced ideal in $W(G)$. Then:

REMARK 4.2. *For Γ a hyperbolic surface group and $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$ in the same Anosov component, then*

$$M_\rho \equiv_{diffeo} M_{\rho'}. \tag{39}$$

4. Proof of Theorem 4.1

PROOF. (Theorem 4.1) Let $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$ irreducible and in the same Anosov component. We want to show the existence of $F : \mathcal{F}^3 \to \mathcal{F}^3$ a topological connjugacy of ρ, ρ'.

Then by Thm. 4.4, there exists a diffeomorphism $D : M_\rho \to M_{\rho'}$. This lifts to a diffeomorphism of the universal cover $\tilde{D} : \tilde{M}_\rho \to \tilde{M}_{\rho'}$. Then by Lemma 4.3.1, the domains of discontinuity Ω_ρ, $\Omega_{\rho'}$ are simply connected, so $\tilde{M}_\rho \equiv \Omega_\rho$ and $\tilde{M}_{\rho'} \equiv \Omega_{\rho'}$. So we can consider $\tilde{D} : \Omega_\rho \to \Omega_{\rho'}$.

Clearly $\tilde{D}$ is a topological conjugacy of Ω_ρ and $\Omega_{\rho'}$, since it is a diffeomorphism and for all $f \in \Omega_\rho$ and $g \in \Gamma$ we have $\tilde{D}(\rho(g) \cdot f) = \rho'(g) \cdot \tilde{D}(f)$.

Next, we want to extend $\tilde{D}$ to a homeomorphism $F : \mathcal{F}^3 \to \mathcal{F}^3$ such that $F|_{\Omega_\rho} \equiv \tilde{D}$. By Theorem 4.3, Ω_ρ is an open, connected dense subset of $\mathcal{F}^3$, so $\tilde{D}$ has a unique continuous extension to $\mathcal{F}^3$ using sequences in Ω_ρ which limit to points in $Th(\Gamma)$.

TODO check thet F is a topological conjugacy. TODO where do we use irreducible? □

Remark 4.3. *The topological conjugacy F may not be unique, however it is uniquely determined by the choice of the diffeomorphism D and its lift $\tilde{D}$.*

Proof. (Lemma 4.3.1) Fact: $\pi_1\mathcal{F}^3$ is trivial. Fact: The odd homology groups $H^{2n+1}(\Omega_\rho)$ vanishes. Fact: Then $\pi_1(\Omega_\rho)$ is perfect, since $\pi_1(\Omega_\rho)^{ab} = H^1(\Omega_\rho) = \{e\}$.

Claim: First consider $\Omega_{\rho_{Fuchs}}$ where ρ_F is $SL_3\mathbb{C}$-Fuchsian. Then $\Omega_{\rho_{Fuchs}}$ is simply connnected.

Claim: If there is a transverse homotopy between $\Lambda_{\rho_{Fuchs}}$ and Λ_ρ, then $\pi_1(\Omega_{\rho_{Fuchs}}) = \pi_1(\Omega_\rho)$.

Claim: There is a transverse homotopy between $\Lambda_{\rho_{Fuchs}}$ and Λ_ρ. □

5. Topological conjugacies of reducible representations

This argument for Theorem 4.2 hinges on whether ρ fixes a line and/or a plane in $\mathbb{C}^3$. Let $\rho : \Gamma \to SL_3\mathbb{C}$ reducible. We say that ρ is

(1) *semisimple* if ρ fixes a line $l \in CP^2$ and a plane $H \in (CP^2)^*$,
(2) *line-reducible* if ρ fixes a line $l \in CP^2$, but no plane,
(3) *plane-reducible* if ρ fixes a plane $H \in (CP^2)^*$, but no line, and
(4) *neither line- nor plane-reducible* otherwise.

Given $l \in \mathbb{CP}^2$, we use the abuse of notation

$$\alpha_1(l) := \alpha_1(l, H) \tag{40}$$

for any $H \supset l$ because $\alpha_1(l, H)$ does not depend on choice of H. Similarly, for $H \in (\mathbb{CP}^2)^*$, we use the analogous abuse of notation

$$\alpha_2(H) := \alpha_2(l, H) \text{ for some } l \subset H. \tag{41}$$

If ρ is semisimple, then for all $x, y, z \in \Lambda_\rho^{(3)}$ we have $x_y = x_z$, $y_x = y_z$, $z_x = z_y$, and $x^y = x^z$, $y^x = y^z$, $z^x = z^y$. So ρ fixes the line $l = l_{x^y}$ and the plane $H = H_{x_y}$. This means that the following sets are invariant under ρ:

$$\alpha_1(l) = span(x_y, y_x) = span(x_z, z_x) = span(y_z, z_y), \text{ and} \tag{42}$$

$$\alpha_2(H) = span(x^y, y^x) = span(x^z, z^x) = span(y^z, z^y). \tag{43}$$

Similarly, if ρ is line-reducible (but not semisimmple), then all $x^y = x^z$, $y^x = y^z$, $z^x = z^y$ but not all $x_y = x_z$. So ρ also fixes the line $l = l_{x^y}$, so the following set is invariant under ρ:

$$\alpha_1(l) = span(x^y, y^x) = span(x^z, z^x) = span(y_z, z_y). \tag{44}$$

If ρ is plane-reducible (but not semisimple), then for all $x, y, z \in \Lambda_\rho^{(3)}$ we have $x_y = x_z$, $y_x = y_z$, $z_y = z_x$, but not all $x^y = x^z$. So ρ fixes the plane $H = H_{x_y}$, so the following set is invariant under ρ:

$$\alpha_2(H) = span(x_y, y_x) = span(x_z, z_x) = span(y_z, z_y). \tag{45}$$

Finally, if ρ is neither line- nor plane-reducible, then ρ fixes no complex line or plane in $\mathbb{C}^3$.

Proof. (Theorem 4.2) If ρ, ρ' are in the same representation conjugacy class and have the same reducible type, then they are topologically conjugate.

Now we consider pairs $\rho, \rho' \in \overline{[\rho]}$ with different reducible types.

Semisimple representations are only topologically conjugate to other semisimple representations.

Since line-reducible and plane-reducible representations but fix a complex line in Ω_ρ, then they can be topologically conjugate.

Finally, if ρ is neither line- nor plane-reducible, then it does not fix any complex line in Ω_ρ, and is therefore not topologically conjugate to other reducible types. However, it is topologically conjugate to other irreducible representations in the same Anosov component. □

We also gain the following corollary, which is immediate from the proof of the Theorem and Lemma 3.1.2:

Corollary 4.4.1. *Suppose F a topological conjugacy between ρ line-reducible and some reducible ρ'. If ρ' line-irreducible, then F matches Richardson varieties. Otherwise ρ' is plane-irreducible and F switches Richardson varieties.*

CHAPTER 5

Vitae

For UIC this can be a full CV or a short version (e.g. half a page) with previoius degrees, work experience and publications.

APPENDIX A

Technical Lemmas

Here we collect some supporting lemmas.

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